



Host-side DAMON Hotness under KVM/THP:

Granularity Gaps and Evidence for Demotion



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A deployment needs THP and active-memory data

Sangfor Technologies: KVM/QEMU with shared tmpfs RAM

THP performance

Enabled by default
Customer workloads vary

Active-memory estimates

Inform tier capacity
Guide placement decisions

Host DAMON reports much more hot memory with THP under the same guest access pattern.



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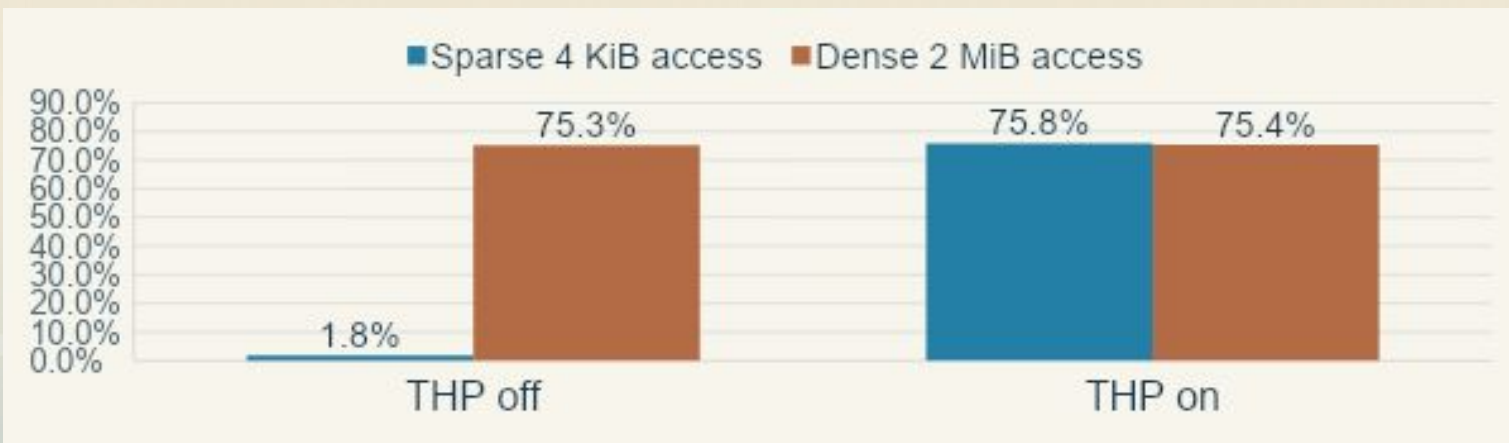
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THP makes sparse and dense activity look similar

Controlled reproduction: 64 GiB VM RAM, 48 GiB resident data



Reported hot bytes / 64 GiB backend



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Community discussion led to broader workloads

masim extensions

- Logical hotsets and phase changes
- Hot-to-cold and hot-to-warm cases
- Per-page reference data for evaluation

Fixed-work THP check

Hotset	Elapsed reduction
64 MiB	3.85%
1 GiB	26.57%

THP on vs off, paired medians
100M writes, 3 repeats per hotset



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TPP-like experiments expose a replacement gap

Two DRAM NUMA nodes, 56 GiB VM, 8 GiB logical near-tier goal

New-hot samples near the CPU at the end of recovery

Repeat	Observe only	Idle demotion	Broader demotion
1	1 / 512	158 / 512	152 / 512
2	0 / 512	135 / 512	154 / 512
3	1 / 512	157 / 512	172 / 512

Old warm samples showed no net exit in all three repeats.



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PMU as an additional source of access evidence

Intel PEBS AMD IBS Arm SPE

Collaboration around Ravi's perf RFC v3 and Kunwu's SPE work

Potential value

Finer address samples
within large pages

The comparison we need

KVM address ownership
Coverage and CPU cost
Changes in placement quality

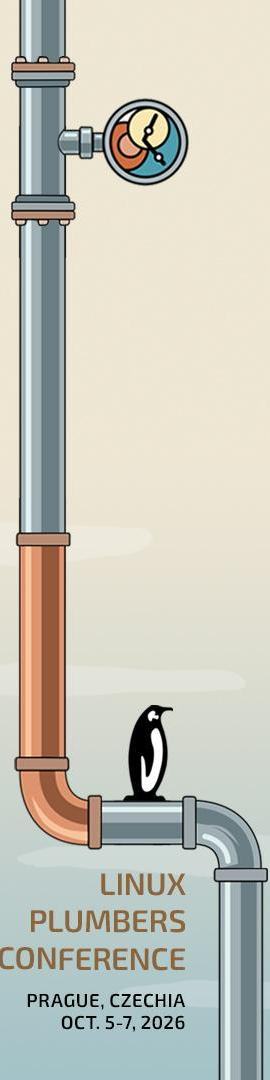
Native PEBS calibration succeeded.
KVM decision benefits still need an end-to-end experiment.



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Next steps and discussion

THP and memory tiering

Separate selection, quota and MM execution costs

DeepSeek sandbox follow-up

Reclaim/reporting granularity and adaptive policies

DAMON extensions

PMU collaboration and follow-up milestones

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